05/09/2024

LSM

Distribution of sample partial correlation coefficient between Y and X_1 given $X_2, \ldots, X_{p-1}$

$$r_{Y X^{(1)} | X_{(-1)}} = \frac{Y^T (I - P_{X_{-1}}) \underline{X}^{(1)}}{\sqrt{Y^T (I - P_{X_{-1}}) Y} \sqrt{\underline{X}^{(1)T} (I - P_{X_{-1}}) \underline{X}^{(1)}}}$$

$$X = [1 \ \underline{X}^{(1)} \ \underline{X}^{(2)} \ \ldots \ \underline{X}^{(p-1)}]$$

$$X_{-1} = [1 \ \underline{X}^{(2)} \ \ldots \ \underline{X}^{(p-1)}]$$

Recall: $$\hat{\beta}_1 = \frac{Y^T (I - P_{X_{-1}}) \underline{X}^{(1)}}{X^{(1)T} (I - P_{X_{-1}}) X^{(1)}} = \frac{\underline{W}^T \underline{Y}}{\underline{W}^T \underline{W}}$$

Suppose $$\underline{W} = (I - P_{X_{-1}}) \underline{X}^{(1)}$$

$$r^2_{Y X^{(1)} | X_{(-1)}} = \frac{Y^T W W^T Y / W^T W}{Y^T (I - P_{X_{-1}}) Y} = \frac{Y^T W (W^T W)^{-1} W^T Y}{Y^T (I - P_{X_{-1}}) Y}$$

$$P_X = P_{X_{-1}} + P_{\underline{X}^{(1)} | X_{-1}}$$

$$= P_{X_{-1}} + P_W$$

$$= \frac{Y^T P_W Y}{Y^T (I - P_X) Y + Y^T P_W Y}$$

$$r^2_{Y X^{(1)} | X_{(-1)}} \overset{D}{=} \frac{\sigma^2 \chi^2_1 (\delta/2)}{\sigma^2 \chi^2_{n-r} + \sigma^2 \chi^2_1 (\delta/2)}$$

$$Y^T P_W Y = \frac{(Y^T W)^2}{W^T W}$$

$$\delta = \frac{\beta_1^2 W^T W}{\sigma^2}$$

$$\hat{\beta}_1 = \frac{Y^T W}{W^T W} \sim N\left(\beta_1, \frac{\sigma^2}{W^T W}\right)$$

If $H_0: \beta_1 = 0 \Rightarrow \delta = 0$

then $r^2_{YX^{(1)}|X_{(-1)}} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-r}{2}\right)$

$$r_{YX^{(1)}|X_{(-1)}} = \frac{Y^T W / \sqrt{W^T W}}{\sqrt{Y^T(I-P_X)Y + Y^T P_W Y}} \qquad P_X = P_{X_{-1}} + P_W$$

$$= \frac{\hat{\beta}_1 \sqrt{W^T W}}{\sqrt{SSE + \hat{\beta}_1^2 W^T W}}$$

$$= \frac{\hat{\beta}_1 / \sqrt{\frac{SSE}{W^T W}}}{\sqrt{1 + \left(\frac{\hat{\beta}_1}{\sqrt{SSE/W^T W}}\right)^2}}$$

$$= \frac{\sqrt{n-r}\, T}{\sqrt{1 + (n-r) T^2}}$$

where $T = \frac{\hat{\beta}_1}{S(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\sqrt{MSE/W^T W}}$

$MSE = \frac{SSE}{n-r}$

$r^2_{YX^{(1)}|X_{(-1)}}$ = coefficient of partial determination b/w Y and $X^{(1)}$ given the rest.

Notice $$r^2_{YX^{(1)}|X_{(-1)}} = \frac{Y^T(P_X - P_{X_{-1}})Y}{Y^T(I - P_{X_{-1}})Y}$$

$$r^2_{YX^{(1)}\ldots X^{(k)}|X^{(k+1)}\ldots X^{(p-1)}}$$

$$= \frac{\underset{\sim}{Y}^T(P_X - P_{X_{-(1:k)}})\underset{\sim}{Y}}{\underset{\sim}{Y}^T(I - P_{X_{-(1:k)}})\underset{\sim}{Y}}$$

$$X_{-(1:k)} = [\underline{1} : \underline{X}^{(k+1)} : \ldots : \underline{X}^{(p-1)}]$$

Exact Linear Dependence vs. Collinearity

$X = [X^{(1)} \ldots X^{(p)}]$

→ Corresponds to $\text{rank}(X) < p$

If we assume $\|X^{(j)}\|_2 = 1 \ \forall j$ then X^TX has trace $= p$

condition number of X^TX

$$= \frac{\lambda_{max}(X^TX)}{\lambda_{min}(X^TX)}$$

$$\frac{\|P_{X_{-1}}X^{(1)}\|^2}{\|X^{(1)}\|^2} \approx 1$$ then $X^{(1)}$ collinear ...

$Y_{1j} = \beta_1 + \varepsilon_{1j}$ $\quad j = 1, \ldots, m_1$

$Y_{2j} = \beta_1 + \beta_2 + \varepsilon_{2j}$ $\quad j = 1, \ldots, m_2$

$Y_{3j} = \beta_1 + \beta_2 + \beta_3 + \varepsilon_{3j}$ $\quad j = 1, \ldots, m_3$

Want ① LSE of $\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix} = \underset{\sim}{\beta}$

② BLUE of β_2

③ Test for $H_0: \beta_1 = \beta_2$ vs. $H_A: \beta_1 \neq \beta_2$

Def. $\mu_1 = \beta_1$

$\mu_2 = \beta_1 + \beta_2$ ← 1-1 transformation (reparametrization)

$\mu_3 = \beta_1 + \beta_2 + \beta_3$

↓ after, design matrix is block diagonal...

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\beta_2 = \mu_2 - \mu_1 \Rightarrow \hat{\beta}_2^{BLUE} = \hat{\mu}_2^{LS} - \hat{\mu}_1^{LS}$$

$$\hat{\mu}_i = \underset{\mu_i}{\operatorname{argmin}} \sum_{j=1}^{m_i} (Y_{ij} - \mu_i)^2 = \overline{Y_{i\cdot}}$$

$H_0: \beta_1 - \beta_2 = 0$ vs. $H_A: \beta_1 - \beta_2 \neq 0$

$$\theta = \beta_1 - \beta_2 = 2\mu_1 - \mu_2 \Rightarrow \hat{\beta}_1 - \hat{\beta}_2 = 2\overline{Y_{1\cdot}} - \overline{Y_{2\cdot}}$$

$$\sim N\left(\beta_1 - \beta_2, \frac{4\sigma^2}{m_1} + \frac{\sigma^2}{m_2}\right)$$

Wald's Test

$$(\hat{\theta} - \underset{0}{\cancel{\theta_0}})^T \left(S^2(\hat{\theta})\right)^{-1} (\hat{\theta} - \underset{0}{\cancel{\theta_0}})$$

$$= \frac{(2\overline{Y_{1\cdot}} - \overline{Y_{2\cdot}})^2}{MSE\left(\frac{4}{m_1} + \frac{4}{m_2}\right)}$$

$$SSE = \sum_i \sum_j (Y_{ij} - \hat{\mu}_i)^2 = \sum_i \sum_j (Y_{ij} - \overline{Y_{i\cdot}})^2$$

$$MSE = \frac{SSE}{\sum_{i=1}^{3} m_i - 3}$$

__Ex__ ANCOVA

$$Y_{ij} = M_i + \beta_1 Z_{ij} + \beta_2 Z_{ij} + \varepsilon_{ij} \qquad \begin{matrix} j=1,\dots,m_i \\ i=1,\dots,r \end{matrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \longleftarrow \text{LSE ?}$$

$$\underline{Y} = \begin{pmatrix} Y_{1\cdot} \\ \vdots \\ Y_{r\cdot} \end{pmatrix} = \overset{=A}{\begin{pmatrix} 1_{m_1} & 0 & 0 & & \\ 0 & 1_{m_2} & 0 & & \\ 0 & 0 & 1_{m_3} & \cdots & \\ & \vdots & & & 1_{m_r} \end{pmatrix}} \begin{pmatrix} M_1 \\ \vdots \\ M_r \end{pmatrix}$$

$$+ \underset{=B}{\begin{pmatrix} Z_{1\cdot} & W_{1\cdot} \\ \vdots & \vdots \\ Z_{r\cdot} & W_{r\cdot} \end{pmatrix}} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \varepsilon$$

Calculate $P_A = A(A^TA)^{-1}A^T$ $\qquad A^TA = \begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_r \end{pmatrix}$

$$= \sum_{j=1}^{r} \frac{1}{m_j} u_j u_j^T \qquad u_j = \begin{pmatrix} 0 \\ 1_{m_j} \\ 0 \end{pmatrix}$$

$$Y_{ij} = \underbrace{\left(M_i + \beta_1 \overline{Z}_{i\cdot} + \beta_2 \overline{W}_{i\cdot}\right)}_{} \longleftarrow \theta_i$$

$$+ \beta_1 (Z_{ij} - \overline{Z}_{i\cdot}) + \beta_2 (W_{ij} - \overline{W}_{i\cdot}) + \varepsilon_{ij}$$

$$\begin{pmatrix} \theta_1 \\ \vdots \\ \theta_r \\ \beta_1 \\ \beta_2 \end{pmatrix} \xleftrightarrow{1-1} \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_r \\ \beta_1 \\ \beta_2 \end{pmatrix}$$